COS 201 — Week 3 Files

All three programs were compiled and tested before being put here. Copy each one into a file with the exact name shown (capital letters matter), inside your week03 folder.

SimpleCalculator.java

```
import java.util.Scanner;

// COS 201 - Week 3 Practical 1: Simple Calculator
// Reads two numbers and displays their sum, difference, product and quotient.
public class SimpleCalculator {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);


        System.out.print("Enter the first number: ");
        double firstNumber = input.nextDouble();


        System.out.print("Enter the second number: ");
        double secondNumber = input.nextDouble();


        double sum = firstNumber + secondNumber;
        double difference = firstNumber - secondNumber;
        double product = firstNumber * secondNumber;


        System.out.printf("Sum: %.2f%n", sum);
        System.out.printf("Difference: %.2f%n", difference);
        System.out.printf("Product: %.2f%n", product);


        // Check for a zero divisor before dividing
        if (secondNumber != 0) {
            double quotient = firstNumber / secondNumber;
            System.out.printf("Quotient: %.2f%n", quotient);
        } else {
            System.out.println("Division is undefined for a zero divisor.");
        }


        input.close();
    }
}
```

AverageCalculator.java

```
import java.util.Scanner;

// COS 201 - Week 3 Practical 2: Student Average Calculator
// Reads a student's name and three scores, then displays the total and average.
public class AverageCalculator {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);


        System.out.print("Enter student name: ");
        String name = input.nextLine();
```

```
        System.out.print("Enter score for Course 1: ");
        double score1 = input.nextDouble();

        System.out.print("Enter score for Course 2: ");
        double score2 = input.nextDouble();

        System.out.print("Enter score for Course 3: ");
        double score3 = input.nextDouble();

        double total = score1 + score2 + score3;
        double average = total / 3.0;

        System.out.println("\n--- STUDENT RESULT ---");
        System.out.println("Student: " + name);
        System.out.printf("Total: %.2f%n", total);
        System.out.printf("Average: %.2f%n", average);

        input.close();
    }
}
```

ItemCost.java

```
import java.util.Scanner;

// COS 201 - Week 3 Practical 3: Total Cost of Items
// Reads an item name, quantity and unit price, then displays the total cost.
public class ItemCost {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter item name: ");
        String itemName = input.nextLine();

        System.out.print("Enter quantity: ");
        int quantity = input.nextInt();

        System.out.print("Enter unit price: ");
        double unitPrice = input.nextDouble();

        double totalCost = quantity * unitPrice;

        System.out.println("\n--- PURCHASE SUMMARY ---");
        System.out.println("Item: " + itemName);
        System.out.println("Quantity: " + quantity);
        System.out.printf("Unit price: %.2f%n", unitPrice);
        System.out.printf("Total cost: %.2f%n", totalCost);

        input.close();
    }
}
```

README.md — Week 3 section

Open your existing README.md and paste this at the bottom, below your Week 1 and Week 2 sections.

```
## Week 3: Variables, Data Types, Expressions and Input/Output

### Objectives
- Declare and initialize Java variables.
- Use appropriate primitive and reference data types.
- Perform arithmetic calculations.
- Read keyboard input using Scanner.
- Display formatted output.
- Compile, test and submit Java programs using GitHub.

### Programs
1. `week03/SimpleCalculator.java`
   - Performs arithmetic calculations on two numbers.
2. `week03/AverageCalculator.java`
   - Calculates the total and average of three scores.
3. `week03/ItemCost.java`
   - Calculates the total cost using quantity and unit price.

### How to Run
Open a terminal in the `week03` directory.

Compile and run the calculator:
- `javac SimpleCalculator.java`
- `java SimpleCalculator`

Compile and run the average calculator:
- `javac AverageCalculator.java`
- `java AverageCalculator`

Compile and run the item-cost program:
- `javac ItemCost.java`
- `java ItemCost`

### Testing
Test each program with different valid inputs and verify
that the results match the expected calculations.
```